



What is the effect of movie's prerelease review score on its ROI?

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Goal: To Evaluate whether the movie's prerelease review scores affects its ROI

Background

Most movies today have pre-premiere screenings attended by film critics who publish their reviews before the movie's official release. These reviews may influence the audience's decision to watch the film, which will directly impact the film's ROI.

Method

334 movies released between 2010-2016 were checked. The data included movie reviews from the Rotten Tomatoes website along with the movies' metadata from IMDb.

We divided the movies into two treatment groups:

1. **Treatment = 0:** scores lower than the median of all aggregated scores.
2. **Treatment = 1:** scores higher than the median of all aggregated scores.

outcome: The movie's ROI

Causal methods

S-Learner 2d+1 – Linear regression (ridge)

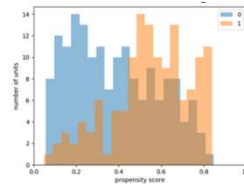
1-NN matching

T-Learner – Linear regression (ridge)

IPW

Assumptions:

- SUTVA
- Common support
- Consistency
- Backdoor criteria



Features used:

- Actors
- Budget
- Genre
- Runtime
- Director
- production company
- Release date

Results

	$\widehat{ATE}$	CI lower bound	CI upper bound	P - value
S-learner	0.5149	-1.0578	2.0877	0.5211
T-learner	0.5761	-1.0575	2.2099	0.4894
IPW	0.259	-1.4616	1.9796	0.7680
1-NN matching	0.8947	-1.0204	2.8099	0.3598

- All ATEs are quite small (less than 1)
- All confidence intervals include 0
- CIs overlap almost completely
- P-values are quite high. (H_0 - No casual relation. $\rightarrow H_1$ - A casual relation exists).
- Can't determine explicitly that there exist a causal relationship

Sensitivity analysis

We measured the model sensitivity to our treatment definition, by binning the review score into more than 2 groups. For each binning method we ran all the causal methods. **The ATE results :**

Number of treatments	S-learner	T-learner	IPW	1-NN matching
2	0.5149	0.5761	0.259	0.8947
3	0.1188	0.2765	-1.067	0.4006
4	-0.5785	-0.4233	-1.197	-0.2097
5	-0.7066	-0.5602	-1.653	-0.0552

Variances:

Number of treatments	S-learner	T-learner	IPW	1-NN matching
3	1.3551	2.1161	0.0117	2.3442
4	2.3607	2.7983	3.3571	2.7439
5	1.646	2.2455	0.8002	0.9906

Conclusions

- It can not be determinate if there is a **causal relationship** between movie's pre-release review scores and their ROIs.
- The effect estimation **is very sensitive** to the treatment definition (binning), one should consider work with causal methods that fit continuous treatments.
- There is a chance SUTVA assumption doesn't hold.

